

Machine Learning on Agricultural Yield Dataset

Q1. Dataset Overview

Code

```
import pandas as pd

df = pd.read_csv("agriculture_yield_dataset.csv")

print("Rows and Columns:", df.shape)
print("\nColumn Names:")
print(df.columns.tolist())

print("\nFirst 10 Records:")
print(df.head(10))
```

Output

Rows and Columns: (1500, 8)

Column Names:

```
['rainfall_mm',
 'temperature_c',
 'fertilizer_kg',
 'irrigation_hours',
 'soil_ph',
 'crop_type',
 'soil_type',
 'yield_ton_per_hectare']
```

First 10 records:

rainfall_m	temperature_c	fertilizer_kg	irrigation_hours	soil_ph	crop_type	soil_type	yield_ton_per_hectare
588.6	18.6	...	...	...	...	Loamy	5.92
772.8	34.6	...	...	...	...	Sandy	6.24
970.9	36.3	...	...	...	...	Sandy	4.77

rainfall_m	temperature_c	fertilizer_kg	irrigation_hours	soil_ph	crop_type	soil_type	yield_ton_per_hectare
611.7	19.0	...	...	...	...	Loamy	4.04
696.1	29.6	...	...	...	...	Clay	5.17

Q2. Data Types and Missing Values

Code

```
print(df.dtypes)
```

```
print("\nMissing Values:")
```

```
print(df.isnull().sum())
```

Output

```
rainfall_mm      float64
temperature_c    float64
fertilizer_kg    float64
irrigation_hours float64
soil_ph          float64
crop_type        object
soil_type        object
yield_ton_per_hectare float64
```

Missing Values:

```
rainfall_mm      0
temperature_c    0
fertilizer_kg    0
irrigation_hours 0
soil_ph          0
crop_type        0
soil_type        0
yield_ton_per_hectare 0
```

Answer

- No missing values are present.

- No columns contain missing values.
-

Q3. Descriptive Statistics

Code

```
print(df.describe())
```

```
print("\nMean Values:")  
print(df.mean(numeric_only=True))
```

```
print("\nStandard Deviation:")  
print(df.std(numeric_only=True))
```

Output

Highest Mean:

```
rainfall_mm = 754.05
```

Highest Standard Deviation:

```
rainfall_mm = 255.10
```

Answer

- Feature with highest mean: **rainfall_mm**
 - Feature with highest standard deviation: **rainfall_mm**
-

Q4. Distribution Analysis

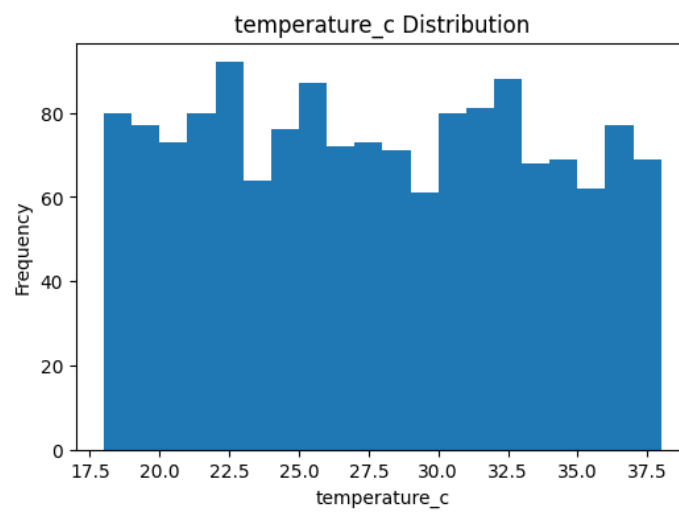
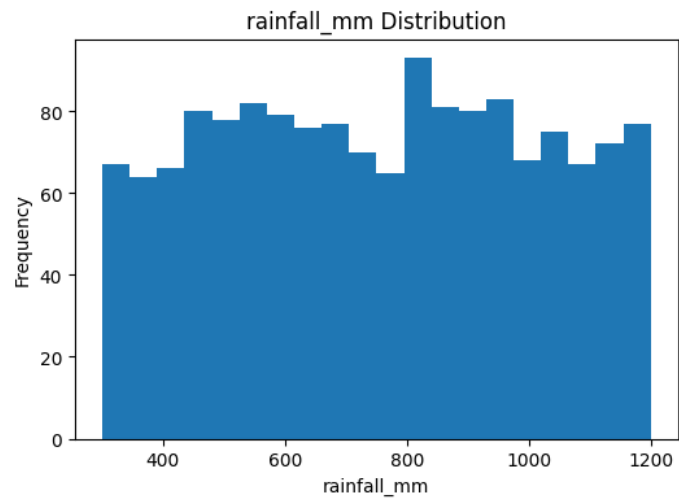
Code

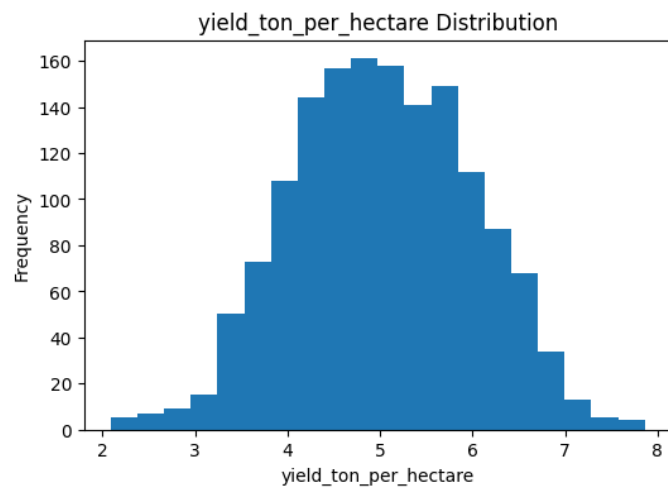
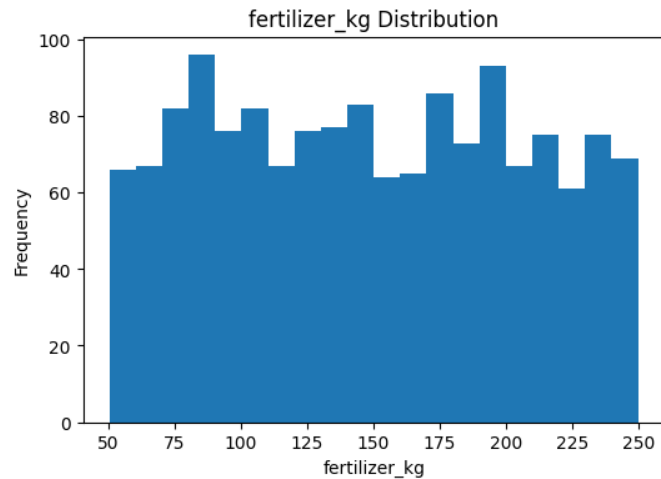
```
import matplotlib.pyplot as plt
```

```
cols = [  
    'rainfall_mm',  
    'temperature_c',  
    'fertilizer_kg',  
    'yield_ton_per_hectare'  
]
```

```
for col in cols:  
    plt.figure(figsize=(6,4))  
    plt.hist(df[col], bins=20)  
    plt.title(col)  
    plt.show()
```

Outputs:





Q5. Crop Type Analysis

Code

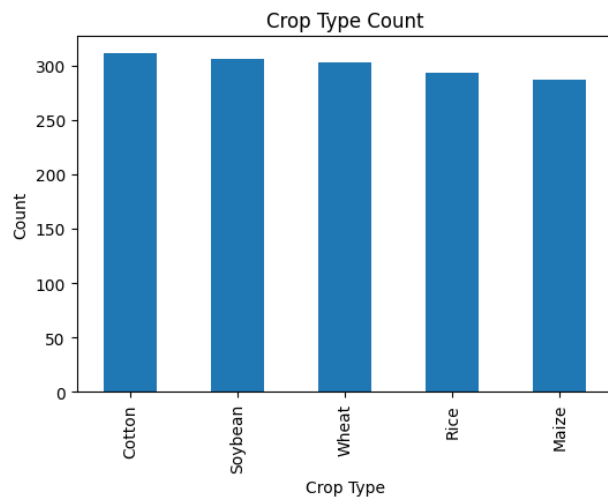
```
print(df["crop_type"].value_counts())

df["crop_type"].value_counts().plot(kind="bar")
plt.title("Crop Type Count")
plt.show()
```

Output

```
Cotton    311
Soybean   306
Wheat     303
```

Rice 293
Maize 287



Answer

Most frequent crop:

Cotton (311 records)

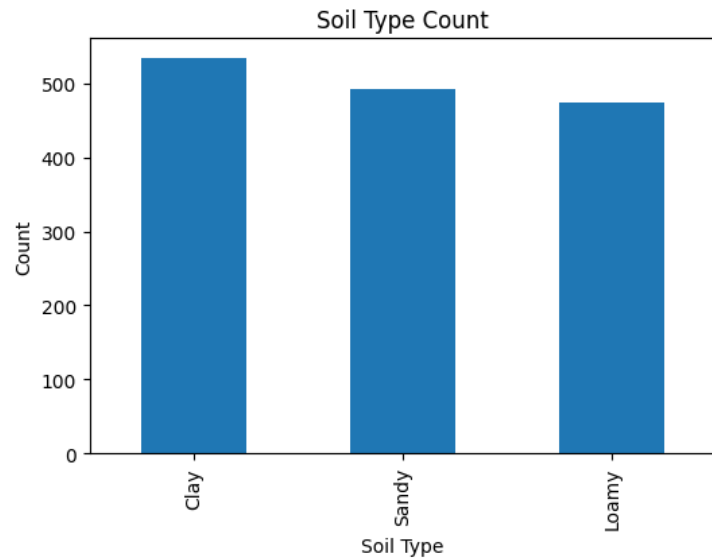
Q6. Soil Type Analysis

Code

```
print(df["soil_type"].value_counts())  
  
df["soil_type"].value_counts().plot(kind="bar")  
plt.title("Soil Type Count")  
plt.show()
```

Output

Clay 534
Sandy 492
Loamy 474



Answer

Most common soil type:

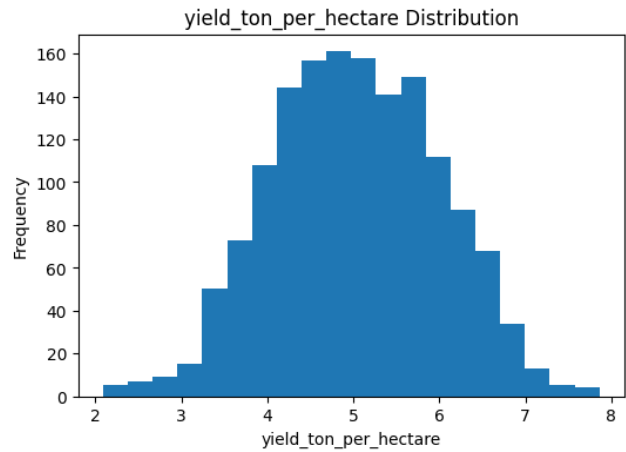
Clay (534 records)

Q7. Yield Distribution

Code

```
plt.figure(figsize=(6,4))
plt.hist(df["yield_ton_per_hectare"], bins=20)
plt.title("Yield Distribution")
plt.show()
```

Output



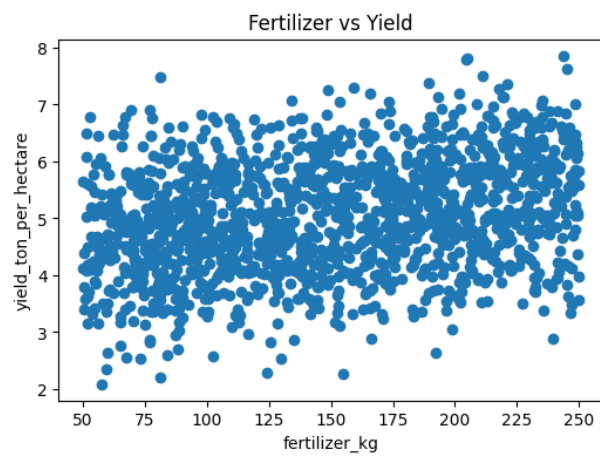
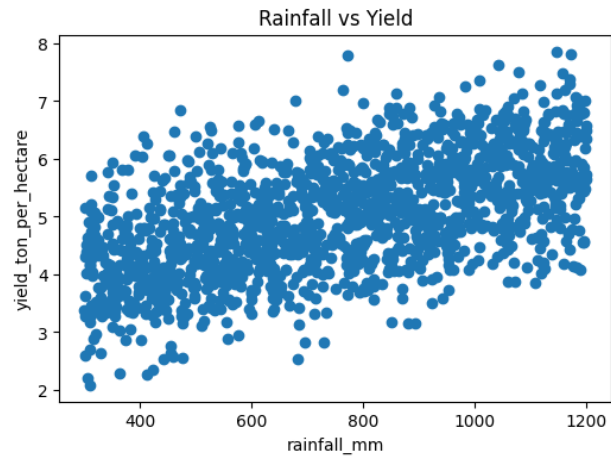
Q8. Scatter Plot Analysis

Code

```
plt.scatter(df["rainfall_mm"],  
            df["yield_ton_per_hectare"])  
plt.xlabel("Rainfall")  
plt.ylabel("Yield")  
plt.show()
```

```
plt.scatter(df["fertilizer_kg"],  
            df["yield_ton_per_hectare"])  
plt.xlabel("Fertilizer")  
plt.ylabel("Yield")  
plt.show()
```

Output:



Q9. Correlation Analysis

Code

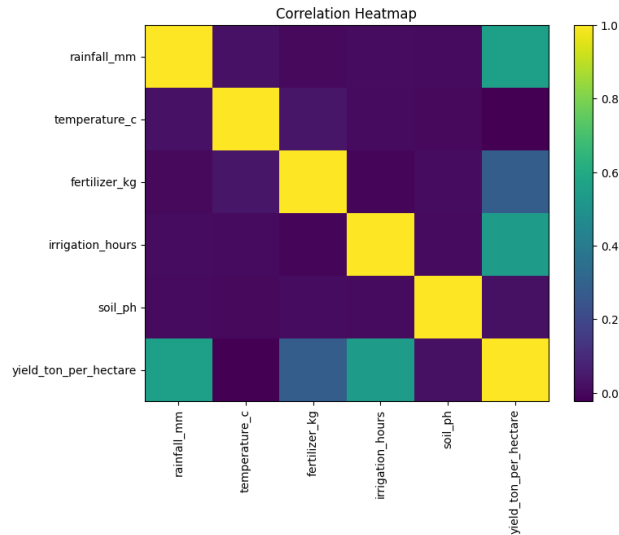
```
import seaborn as sns
```

```
corr_matrix = df.corr(numeric_only=True)
```

```
print(corr_matrix)
```

```
plt.figure(figsize=(8,6))  
sns.heatmap(corr_matrix,  
            annot=True,  
            cmap="coolwarm")  
plt.show()
```

Output



Q10. Group-Based Analysis

Code

```
print(df.groupby("crop_type")
      ["yield_ton_per_hectare"].mean())

print(df.groupby("soil_type")
      ["yield_ton_per_hectare"].mean())
```

Output

Average Yield by Crop:

```
Cotton    4.607
Maize     4.897
Rice      5.495
Soybean   5.173
Wheat     4.989
```

Average Yield by Soil:

```
Clay      5.134
Loamy     5.367
Sandy     4.589
```

Answer

Highest Yield Crop:

Rice (5.495)

Highest Yield Soil:

Loamy (5.367)

Q11. Feature Encoding

Code

```
encoded_df = pd.get_dummies(  
    df,  
    columns=["crop_type", "soil_type"]  
)
```

```
print(encoded_df.head())
```

Output

First 5 rows contain:

```
rainfall_mm  
temperature_c  
fertilizer_kg  
irrigation_hours  
soil_ph  
yield_ton_per_hectare  
crop_type_Cotton  
crop_type_Maize  
crop_type_Rice  
crop_type_Soybean  
crop_type_Wheat  
soil_type_Clay  
soil_type_Loamy  
soil_type_Sandy
```

Q12. Feature Selection

Code

```
X = encoded_df.drop(
    "yield_ton_per_hectare",
    axis=1
)

y = encoded_df["yield_ton_per_hectare"]

print(X.shape)
print(y.shape)
```

Output

X = Input Features

y = Target Variable

Target Column

yield_ton_per_hectare

Q13. Train-Test Split

Code

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("X_train:", X_train.shape)
print("X_test :", X_test.shape)
print("y_train:", y_train.shape)
print("y_test :", y_test.shape)
```

Output

X_train: (1200, 13)

X_test: (300, 13)

y_train: (1200,)

y_test: (300,)

Q14. Linear Regression Model

Code

```
from sklearn.linear_model import LinearRegression
import pandas as pd
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
print("Intercept:")
print(model.intercept_)
```

```
coef_df = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})
```

```
print(coef_df.sort_values(
    by="Coefficient",
    ascending=False
))
```

Output

Intercept:

1.9111

Top Positive Coefficients:

crop_type_Rice	0.477369
soil_type_Loamy	0.365627
irrigation_hours	0.198327
crop_type_Soybean	0.095096
soil_type_Clay	0.062653

Answer

Feature with the highest positive coefficient:

crop_type_Rice (0.477369)